

Transforming Production Planning Through Data Intelligence

Reflection document

Egemen Alkan
Student Bachelor Applied Computer Science

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1. Introduction

This document serves as a comprehensive reflection on my internship project at Estée Lauder, titled Transforming Production Planning Through Data Intelligence. The project was designed to address a critical operational bottleneck: the reactive and manual nature of weekly planning meetings, where planners spent excessive time verifying data rather than making strategic decisions.

My internship focused on two distinct but complementary workstreams:

1. **Primary Deliverable:** Developing an enhanced Power BI report to serve as an automated violation tracker.
2. **Secondary Deliverable:** Conducting a feasibility study and developing a prototype for an AI Planning Assistant.

In the following sections, I will first provide a substantive reflection on the project outcomes and their business value. Subsequently, I will offer a personal reflection on my professional growth, the competencies I addressed, and the challenges I overcame during this transition from student to data professional.

2. Substantive Reflection: The Internship Project

2.1. Accomplishments & Deliverables

My primary goal was to move the planning process from manual data compilation to automated intelligence. I successfully delivered two robust solutions that achieved this:

- **The Automated Violation Tracker (Power BI):** I transformed the existing reporting environment into an active management tool. Specifically, I translated abstract business rules into complex DAX logic. This included sophisticated calculations for **Context-Aware Changeovers** (Rule 8), where the definition of a changeover changes based on the resource type (Tubes vs. Vials) . I also implemented a **Two-Step Duplicate Check** (Rule 10) that intelligently filters out intentional large orders before flagging potential inefficiencies . Furthermore, I integrated a custom **Power App**, allowing planners to write back data and Mark Violations as Agreed directly within the dashboard.
- **The AI Planning Assistant (Copilot Studio):** I went beyond a theoretical research paper. I delivered a formal go recommendation for Microsoft Copilot Studio after proving that OpenAI was non-compliant with enterprise security standards . I then built a high-fidelity prototype featuring a **Multi-Agent Architecture** that successfully routes user intents between live Power BI data queries and SharePoint procedural documents .

2.2. Business Value & User Impact

The impact on the client's organization has been immediate and tangible:

- **Transformation of Meetings:** The weekly meeting has shifted from a data verification session to a strategic decision-making forum. The automated checking of 10 planning rules eliminates hours of manual pre-meeting compilation.
- **Workflow Efficiency:** The integration of the Power App addressed a major user pain point. By creating a persistent log of agreed violations, I prevented the team from wasting time discussing the same issues week after week.

- **Democratization of Data:** The AI prototype demonstrated that complex data (like Missing Parts or Ready to Firm orders) can be accessed via natural language, reducing the barrier to entry for new planners and speeding up information retrieval .

2.3. Project Status: Finished vs. Future Work

- **Power BI Report:** This project is finished. It is fully automated with daily refreshes. However, regarding the original scope of 10 standard rules, Rules 4 (Planning Wheel) and 9 (Planning Frequency) were effectively descope. This was not a failure of implementation but a data readiness decision; the underlying sequence data required to calculate these rules reliably did not exist in the model.
- **AI Assistant:** The feasibility study and prototyping phase are complete. The prototype achieved a 100% success rate across all benchmark KPIs. It is currently a validated prototype ready for further production and a phased rollout to Microsoft Teams.

2.4. Advice for the Client

Based on my development experience, I have two key recommendations for the future:

1. **Data Maturity:** To implement the descope Planning Frequency rule (Rule 9), the business must first formalize the Planning Wheel sequence in the master data. Once this data is available, the logic I developed for the other rules can be adapted.
2. **Scaling the AI:** The AI agent is currently scoped to Missing Parts and Ready to Firm data to ensure high accuracy. I recommend scaling this by creating additional child agents for other production tables, slowly expanding the schema coverage while maintaining the multi-agent architecture.

3. Personal Reflection: Professional Growth

3.1. What the Internship Meant to Me

This internship was the bridge between academic theory and professional reality. It taught me that in a business context, a correct code solution is not enough; it must also be a compliant and usable solution. Seeing the Plant Manager applaud the AI demonstration gave me immense confidence in my ability to deliver value at a high level.

3.2. Competencies and Skills Acquired

- **Advanced DAX & Data Modelling:** I grew significantly in my technical ability. I moved beyond simple aggregations to writing complex DAX using virtual tables to solve the chronological sorting issues in the Changeover rule. I learned to navigate data model ambiguity where standard relationships failed.
- **AI Architecture & Engineering:** I learned that AI requires rigorous engineering. I mastered the concept of Prompt Engineering by embedding database schemas directly into agent instructions to enable accurate DAX generation. I also learned to build Multi-Agent Systems to overcome context limitations.
- **User-Centred Design:** Initially, I built a dashboard crowded with detailed tables. Feedback from the team taught me that less is more. I learned to prioritize a triage view (summary cards) to fit the 30-minute meeting constraint, only showing details when necessary .

3.3. Problems Faced and Solutions

I encountered two major challenges that defined my learning experience:

3.3.1. The Data Ambiguity Challenge (Technical)

- Problem: Early calculations for Rule 8 (Changeovers) failed because a single material could be linked to multiple setup groups, causing standard DAX relationships to break.
- Solution: I realized I couldn't rely on the static model. I wrote a measure that builds a virtual table in memory to rank orders chronologically and compare them row-by-row. This taught me the power of handling context programmatically rather than relying solely on the data model .

3.3.2. The Enterprise Security Challenge (Strategic)

- Problem: I initially explored OpenAI's Agent Builder for the AI Assistant because it was low-code and easy to use. However, my research revealed it could not handle Entra ID authentication or keep data within the tenant, violating company policy .
- Solution: I had to pivot the entire strategy. I conducted a comparative analysis and recommended Microsoft Copilot Studio. Although it required a steeper learning curve to build the multi-agent architecture, it was the only compliant path. This taught me that in an enterprise, security and governance are non-negotiable requirements for software architecture.

4. Conclusion

My internship at Estée Lauder has been a transformative experience. I succeeded in delivering a dual-value project: an immediate operational fix via the Power BI Deviation Tracker that saves time today, and a strategic roadmap via the AI Planning Assistant that prepares the team for tomorrow.

I am leaving the organization not just with a set of finished files, but having deployed a live tool that fundamental changes how the planning team operates. Personally, I have evolved from a student developer into a data consultant capable of balancing technical perfection with business practicality.